

1.1.1 & 1.1.2 Discrete vs Continuous & Digital Systems

منفصل (Discrete) مقادیر وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔

CS XI

Chapter 1: Computer Systems

Lecture 01 of 12

1.1.1 & 1.1.2 Discrete vs Continuous & Digital Systems

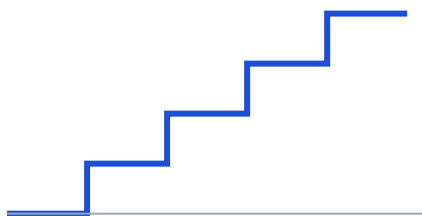
LEARN IT

Everything around you that can be measured or counted is a **quantity**. Quantities come in two kinds.

Discrete quantities are separate, countable items with specific values. You can count them one by one: 1, 2, 3 ... There is nothing "in between". You cannot have 2.5 students in a classroom.

Continuous quantities can take *any* value within a range. They flow smoothly without breaks. Water can be 30 °C, 30.5 °C, 30.51 °C, and so on — there is always another value in between.

Discrete (counted)	Continuous (measured)
Number of students in a classroom	Temperature of water
Number of cars in a parking lot	Speed of a moving car
Has fixed, separate values	Has any value in a range



Discrete (Stairs)



Continuous (Ramp)

Stairs have fixed steps (discrete). A ramp lets you stand at any point (continuous).

REMEMBER IT

Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask yourself: "Do I *count* it or *measure* it?" Count → discrete. Measure → continuous.

COMMON MISTAKE

Students sometimes say "money is continuous because amounts are large". No — rupees and paisa are counted in fixed units, so money is discrete. Size does not matter; what matters is whether values in between are possible.

اردو میں سمجھیں

منفصل (Discrete) مقداریں وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔ مسلسل (Continuous) مقداریں کسی حد کے اندر کوئی بھی قیمت لے سکتی ہیں، جیسے پانی کا درجہ حرارت۔ سیڑھیاں منفصل ڈیٹا کی مثال ہیں اور ڈھلوان (ریمپ) مسلسل ڈیٹا کی۔

CHECK YOURSELF

1. Is the number of pages in a book discrete or continuous?
2. Is your height discrete or continuous?
3. In the stairs/ramp analogy, which one shows continuous data?

Answers: 1. Discrete (you count pages). 2. Continuous (height is measured and can be any value). 3. The ramp.

LEARN IT

A **digital system** is an electronic system that uses only two values: **0** and **1**. These two values are called **binary digits**, or **bits** for short.

- **0** represents OFF, LOW, or FALSE.
- **1** represents ON, HIGH, or TRUE.

Why only two values? Because it is easy for an electronic circuit to tell "voltage present" from "no voltage". Every number, letter, photo and song in a computer is finally stored as a long pattern of these bits.

Advantages of digital systems

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Examples of digital systems

- **Computers:** process data as 0s and 1s.
- **Smartphones:** store photos and music as binary data.
- **Digital cameras:** capture images as binary code.

WORKED EXAMPLE

Why is copying digital data "accurate"?

Step 1: A song is stored as bits, for example 10110010...

Step 2: When you copy it to a USB, the computer copies each bit exactly: 1 stays 1, 0 stays 0.

Step 3: Even after 100 copies the bits are identical, so the song sounds the same. An old cassette tape (analog) loses a little quality every time it is copied.

REMEMBER IT

Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

اردو میں سمجھیں

ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بیت (Bit) کہتے ہیں۔ 0 کا مطلب بند، کم یا غلط ہے اور 1 کا مطلب چالو، زیادہ یا درست ہے۔ ڈیجیٹل نظام قابل بھروسہ، درست، بنانے میں آسان اور سافٹ ویئر سے قابو میں رہنے والے ہوتے ہیں۔

CHECK YOURSELF

1. What does a 1 represent in a digital system?
2. Give two advantages of digital systems.

Answers: 1. ON, HIGH or TRUE. 2. Any two of: reliable (less affected by noise), accurate (no loss when copying), easy to design (two states), programmable (controlled by software).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A discrete quantity is:

- a. Counted in separate steps
- b. Any value on a smooth range
- c. Always analog
- d. A sine wave

MCQ 2. Temperature of tea is:

- a. Continuous
- b. Discrete
- c. Only Boolean
- d. A gate

MCQ 3. Number of students in a class is:

- a. Discrete
- b. Continuous
- c. Analog voltage
- d. A ramp

MCQ 4. The stairs vs ramp analogy: stairs are:

- a. Discrete
- b. Continuous
- c. TCP
- d. Agile

MCQ 5. Digital systems use:

- a. 0 and 1
- b. Every real number
- c. Only sine waves
- d. Paper only

MCQ 6. A bit is:

- a. A binary digit, 0 or 1
- b. Eight bytes
- c. A logic gate
- d. A database

Extra short questions

1. Differentiate discrete and continuous with one example each.
2. Why do computers prefer digital values?
3. What is a digital system?
4. Give two advantages of digital systems.

Extra long question

1. With a labelled stairs/ramp sketch, explain discrete vs continuous and link both to analog vs digital signals.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Discrete: countable steps (cars in a park). Continuous: any value in a range (speed of a car).
2. Bits 0/1 can be restored after noise; analog waves are bent by interference.
3. An electronic system that stores and processes only two values, 0 and 1.
4. Reliable (noise resistant), accurate copies, easy to design (two states), programmable.

Long

1. Stairs = discrete/digital (fixed levels). Ramp = continuous/analog (smooth). Computers convert analog measurements into bits, process bits, then convert back if a human must see the result.

Next lecture: 1.1.3 Analog and Digital Signals